Project instructions.

This is a guide for your course project in MTH4322 of fall 2026, with Professor Adam Sheffer.

The project should be done using an AI of your choice. When done, you should send Adam:

- The project itself. If possible, the recommended option is to place it on your GitHub. But an email is also fine.
- An initial design document (see below for more details).
- The chat log with AI together with a detailed explanation of the checks you ran at each stage. Screenshots would be great!
- List the AI system(s) used, the programming language(s) used, and any other technology used.
- Any other information that might be relevant: Comments, unusual stories, bad jokes, etc.

All prompts must be written by you. You may **not** use any AI to generate, rewrite, or improve your prompts. You may use AI to help create testing tools or test data. You may also ask it to suggest small examples, but you must independently determine or verify the correct answers. You must perform and understand the verification yourself.

While AI may handle the algorithm, implementation details, and coding, you must understand everything in the project. You may not just assume that something is okay without understanding it. (You do not need to manually verify every line of code, but you should understand the purpose and structure of every major part of the program.)

When starting the project, before using AI, first create a general design of what you want the system to look like to users. This part does not need to include any algorithmic aspects. Think about how the system would look nice to the user and be easy to use.

Use AI to create the system in steps. Each time, add another part and carefully test that it works. You are expected to be skeptical about everything. **Check everything, ask why things work, and be suspicious**. More generally, it is common to test three types of cases:

1. Standard inputs, especially small inputs where you can determine the correct answer yourself.
2. Edge cases.
3. Randomly produced cases.

Checking complicated systems with large inputs is tricky. Use AI to create tools or pieces of data to help with that. For example, you might like to create a second variant of your

program with random input already in it. Did your tests lead you to change the system design? Did they expose bugs?

While you need to understand the algorithm and running time, there is no need to submit that. Adam will ask about those in class, so please be ready to discuss that. Adam may also ask how various changes to the problem may affect the algorithm.

Some projects involve problems for which libraries already exist that directly implement the required algorithm. You **may not** use such libraries unless explicitly permitted. Ordinary libraries for user interfaces, data structures, graphics, networking, etc are fine.